

Anil Kumar Kakarla

(Sr. DevOps Engineer)

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ABOUT ME: Senior Cloud DevOps Engineer with over 11 years of experience in IT, specializing in building, deploying, and managing cloud-based infrastructures. With more than six years of hands-on expertise in AWS and Azure, I have a proven track record in automating workflows using tools like Terraform, Ansible, and Jenkins. I am skilled in containerization technologies such as Docker and Kubernetes, focus on ensuring environment security, high availability, scalability, and performance optimization. Adept at designing and optimizing CI/CD pipelines to accelerate software delivery and streamline operations in cloud environments.

PROFESSIONAL SUMMARY:

- Extensive experience with AWS (EC2, S3, RDS, EKS, VPC etc.), Azure (Blob Storage, Virtual Network, Load Balancer, AKS). Proficient in multi-cloud environments.
- Developed and deployed cloud-native applications using serverless technologies like AWS Lambda and Azure Functions.
- Managed secure cloud networking, VPN setups, VPC peering, load balancing, and security best practices, including IAM, Azure AD and firewall management.
- Familiarity with software engineering best practices, including unit testing, version control, continuous delivery, disaster recovery, and incident management.
- Strong in Shell scripting, Python, YAML for automation, configuration, and DevOps workflows.
- Created and managed infrastructure as code (IaC) using Terraform to automate the provisioning of cloud resources across AWS and Azure.
- Expertise in managing cloud infrastructure with Terraform for automating provisioning, configuration, and deployment of cloud resources.
- Proficient with Git for version control, branching strategies, and GitOps practices for infrastructure management and continuous delivery.
- Implemented version-controlled infrastructure workflows with Git, CI/CD pipelines, and Terraform Cloud.
- Used Terraform state management strategies (remote state backends, locking) to ensure collaboration and state consistency across teams.
- Managed branching strategies like GitFlow, creating feature branches for development, merging into the develop branch for integration, and then into main/master for production, ensuring a smooth workflow from development to deployment.
- Implemented the Shift-Left approach by integrating automated testing and security checks early in the development lifecycle, using tools like SonarQube and Veracode. This reduced defects and minimized production failures, ensuring higher code quality.
- Designed and maintained CI/CD pipelines using Jenkins, Azure DevOps Pipelines for automated build, test, and deployment.
- Created and managed Jenkins pipelines using its master-agent architecture to automate build, test, and deployment processes. Experienced in setting up distributed builds, enabling efficient execution across multiple environments.
- Managed and deployed applications in production using blue-green and canary deployment strategies. Experienced in automating release processes.
- Implemented DevOps practices using tools like Nexus, JFrog Artifactory, and managed CI/CD pipelines integrated with Docker, Kubernetes, and Terraform.
- Implemented monitoring solutions with Prometheus, Grafana, ELK Stack, AWS CloudWatch, Azure Monitor, Splunk to ensure infrastructure performance and application health.
- Expertise in Linux/Unix system administration (RedHat, Ubuntu, CentOS), server builds, upgrades, migrations, and troubleshooting.

- Managed databases on AWS RDS, Azure SQL, MySQL, MongoDB optimizing performance, monitoring, and security.
- Strong proficiency in Docker and Kubernetes (AKS, EKS, GKE), including managing clusters, Helm charts, and deploying microservices in scalable environments.
- Created and managed Kubernetes clusters for orchestrating containerized applications, ensuring high availability and scalability. Experienced in deploying and scaling applications using Kubernetes, along with managing Helm charts for reproducible builds.
- Created Docker images for containerized applications and managed their deployment across different environments. Used Docker to ensure consistency and portability of applications from development to production.
- Wrote modular Terraform code to promote reusability and scalability across environments.

TECHNICAL SKILLS:

OS Administration	RHEL 7.x/6.x/, Centos, Ubuntu, Windows Server 2008 –R2/2012-R2
Configuration management	Ansible, Salt-Stack
Scripting Language	Bash, Python, Powershell
Application/Web Servers	Tomcat, Apache, NodeJS,glassfish
Virtualization Platforms	Virtual Box, Vagrant, Docker
Infrastructure as A Service	AWS, Microsoft Azure
Logging	Splunk, Datadog & ELK.
Bug Tracking Tools	Service NOW, JIRA, CONFLUENCE
Operating Systems	Windows, Linux, Unix, Mac OS
Databases	MongoDb, Elasticsearch, Redis, mysql
OS Administration	RHEL 7.x/6.x/, Centos, Ubuntu, Windows Server 2008 –R2/2012-R2
Configuration management	Ansible, Salt-Stack
Scripting Language	Bash, Python, Powershell
Application/Web Servers	Tomcat, Apache, NodeJS,glassfish
Virtualization Platforms	Virtual Box, Vagrant, Docker
Infrastructure as A Service	AWS, Microsoft Azure

CERTIFICATION

- AWS Certified Solutions Architect Associate
- RHCSA (Redhat Certified System Administrator)
- RHCE (Redhat Certified System Engineer)

WORK EXPERIENCE:

PWC, USA.
Sr. DevOps Engineer.

Oct 2022 to present.

Project: Supported Quick Suite, a comprehensive framework designed to streamline application development, deployment, and testing. The framework enables rapid development of mobile and desktop applications using Angular/Ionic and Node.js, adhering to modern design principles, professional best practices, and CI/CD readiness. Quick Suite also provides a pre-configured CI/CD pipeline with shared Kubernetes infrastructure to automate workflows and

features functional testing automation for desktop, web, mobile, and microservices applications, ensuring efficiency and consistency across the development lifecycle.

Responsibilities:

- Develop CICD pipelines for Deployments and Troubleshooting the build failures in Azure devops for Java, NodeJS, Python.
- Used Azure pipelines to drive all micro services builds out to the JFOG registry and then deployed Kubernetes, Created Pods, Deployments, Replicas and managed using Kubernetes.
- Expertly utilized Docker for containerizing applications and Kubernetes for orchestration on Azure AKS, enhancing deployment efficiency and application scalability.
- Designed and implemented CI/CD pipelines in Azure DevOps, reducing deployment times and improving deployment frequency.
- Collaborate with Operations, Security, and Support teams to resolve connectivity issues and ensure compliance with release management standards.
- Monitoring SLIs, SLOs, and SLAs to ensure system reliability and availability. Experienced in optimizing MTTD, MTTA, and MTTR by implementing proactive monitoring, automated incident response, and resilient infrastructure strategies.
- Created automated pod restart jobs using python script across all clusters, reducing memory and CPU usage spikes by 20%.
- Install and configure the dependence for the application team.
- Firewall request to open ports for services
- Raising and implementing RFC for sprint release.
- Monitoring application specific jobs, pods, services and server health using Datadog
- Creating Datadog dashboards for Monitoring of AKS and GKE clusters
- Working on Service now as a Change Management/ Work Management Agile Tool
- Upgrading AKS, GKE clusters and rancher
- Managing and renewing SSL certificates using rancher.
- Implemented Kubernetes multi-cluster management strategies using tools such as Rancher.
- Implemented Kubernetes security best practices, including role-based access control (RBAC), Network Policies, and Pod Security Policies
- Orchestrated release management across multiple environments (Dev, QA, Prod) using Azure DevOps Releases, ensuring consistent deployment processes and rollback mechanisms.
- Optimized AKS performance and costs by implementing right-sizing node pools, cluster autoscaling, node pooling, HPA, and resource requests with turbonomics tools.
- Built a Git Ops pipeline using **Flux CD** for continuous delivery of microservices in Kubernetes clusters.
- Used **Kustomize** to manage and deploy environment-specific configurations for multi-region Kubernetes clusters.
- Designed and implemented **Blue/Green & Canary deployment pipelines** for production workloads using tools like Kubernetes.
- Leveraged Terraform for infrastructure provisioning and management, allowing for consistent and repeatable deployments across multiple environments.
- Deploy infrastructure using Terraform, including Traffic Manager, automating provisioning and ensuring consistent environment configurations.
- Collaborated with cross-functional teams to define Kubernetes architecture and deploy microservices efficiently.
- Provided guidance and training to junior team members on best practices for Kubernetes and containerization.

Environments:

Azure DevOps, Flux CD, JFrog Artifactory, ServiceNow, Docker, Kubernetes, AKS, GKE, Rancher,

Kustomize, Terraform, Datadog, Shell, python, Blue/Green Deployments, Canary Deployments, Turbonomics, HPA, Cluster Autoscaler, Traffic Manager Load Balancer.

Big Basket
DevOps Engineer SRE

SEP 2021 to OCT 2022 - Bangalore, India

Project: The Instant is a patented, unmanned cashless smart store i.e. stocked with a variety of high-quality snacks, vegetables and groceries. The smart stores are deployed at various locations and consumers connect by using the app the consumers connect to the Instant Cloud network that further enables and allows the connectivity, purchase of products and different purchase options.

Responsibilities:

- Responsible for performing Application Health Check and troubleshooting within the SLA and escalating incidents when required.
- Scrutinizing the production level Transactions and job fails and screening out the root cause by means of monitoring tools.
- Implemented an automated CI/CD pipeline using Jenkins to trigger deployments on AWS EC2 instances, streamlining the software release process. Configured Jenkins pipelines to build, test, and deploy applications by integrating with Git repositories.
- Created multiple monitoring and alerting Dashboards in Grafana for our AWS Resources.
- Involved with the team members, development & operation team on a regular interval to analyze the project needs and employed the planned strategy in the evolution of design and implementation of BB instant project.
- Setting up email alerts to concern support teams.
- Created Python scripts for **performance monitoring, and alerting**, reducing manual intervention.
- Deploying required corncobs to all servers with salt stack
- Responsible to deploy all new innovations and added features to the production across all servers and keeping a track of its functionality usability, and impacts Demonstrate robust troubleshooting skills, the availability to manage multiple cases ensuring timely resolution and high level of satisfaction
- Responsible for providing support in L2/L3 level by coordinating with concerned teams Support the team-building aspects and provide support and guidance to fellow team members when required.
- Seek advice in order to make decisions on complex business issues Provide after-hours support for Infrastructure-related emergencies as well as occasional weekend maintenance.
- Coordinating with Dev/QA teams for customer related issues.
- **Environments:**

Ubuntu, ELK stack, salt stack, python, shell scripting, AWS, Jenkins, Kibana.

Sprinklr, Bangalore, India.
DevOps Engineer SRE

May 2017 – Aug 2021

Project: Sprinklr is the most complete social media management system for the enterprise. We help the world's largest brands do marketing, advertising, care, sales, research, and commerce on Facebook, Twitter, LinkedIn, and 21 other channels globally all on one integrated platform. This project involves maintaining end-to-end infrastructure support

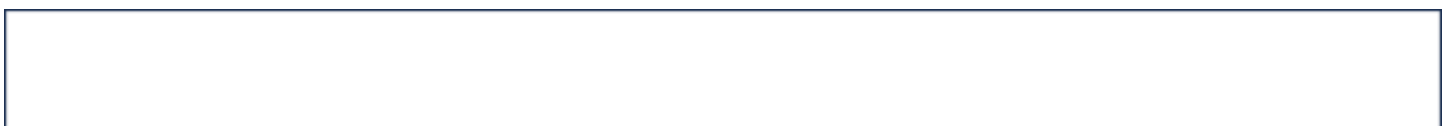
globally and ensures platform availability. Managing infrastructure and automation build deployment and releases. Scaling up infrastructure vertically and horizontally depending on the need Proactively Monitor and fixing issues to ensure availability within SLA's.

Responsibilities:

- Working for the DevOps team, responsible for specialization areas related to Ansible for Cloud Automation. high skills on providing RCA and setting up alert thresholds and Managed SRE team as Lead.
- Compilation and run time issues in Build failures, identifying build breaks and reporting to the development team and interacting with developers to sort issues to build automation using Jenkins.
- Deploy Ansible and Jenkins to complete provision and manage AWS EC2 instances, volumes, DNS and S3.
- Utilized Terraform for automating AWS infrastructure setup, including EC2 instances, S3 buckets, VPCs, and NAT configurations.
- AWS CSA Associate - Involved in designing and deploying multitude applications utilizing AWS stack (Including EC2, VPC, Route53, S3, IAM) focusing on high-availability, fault tolerance and auto-scaling.
- Works with Ansible-playbooks, helm-charts for configuration automation.
- **KAFKA:** Working on Apache Kafka Message Queuing services and looking into queues processing from both producer and consumers, Cluster Management, Disk issues on cluster.
- **ELK:** Experienced in Elasticsearch backend operations (Admin) i.e., technical issues on Elastic Search (Health, create indexes, adding nodes, disable/enable sharding etc.), Logstash and Kibana Management.
- **Release Management:** CI/CD pipeline tools Jenkins. Build and deployments under different testing environments starting from development to. Staging and production.
- **Web Services:** Worked on Glassfish, Tomcat, Java Applications fix Heap, OOM issues, take heap dumps, Jstack and configure java settings accordingly.
- Mesos for Job Scheduling and Maintaining the cluster on Kubernetes.
- Worked on Kubernetes to manage containerized applications using its nodes, Config Maps, selector, Services, and deployed application containers as Pods.
- Experienced in cloud computing technologies like AWS and Azure. Version-control system etc.,
- I have experienced MongoDB, MySQL to create entries, modify, delete and reporting DB issues like query lag, Switch Down and other issues to dev.
- Sound understanding of Networking, Database, Message Queueing technologies.
- Worked on JIRA & Service now as a Change Management/ Work Management Agile Tool.
- Automated NOC alert escalation process with the help of tools such as Prometheus and Grafana.
- Handling Incident/Problem/Change requests through Service-NOW and Jira and day-to-day user administration like adding or deleting users, password aging.

- **Environments:**

Ansible, Terraform, Helm, Jenkins, Git, Mesos, AWS (EC2, VPC, Route53, S3, IAM), Azure, Docker, Kubernetes, Prometheus, Grafana, Apache Kafka, ELK Stack (Elasticsearch, Logstash, Kibana), MongoDB, MySQL, Glassfish, Tomcat, Java (Heap dumps, jstack, OOM handling), DNS, NAT, Load Balancers, ServiceNow, JIRA.



Harman Connected services, Bangalore, India.
Development Engineer

April 2015 - Feb 2017

Project:

Dodge Project involves providing 24X7 support for the critical applications. The objective is to maintain the availability and performance of all these core apps. This project is aimed at migrating total in-house infra/applications to AWS – cloud. The availability of the sites and critical applications are achieved through monitoring the application and web servers and critical applications using monitoring tools such as Nagios. Project scope involves environment monitoring, L1/L2 application support, incident management, change management, configuration management, release management.

Responsibilities:

- Building Linux servers in AWS Adding servers for Nagios monitoring. Setting up OS level checks like Disk checks, CPU Load, CPU utilization Responsible for distributed applications across hybrid AWS and physical data centers.
- Performing Build and Patch Deployment on Test and Production Servers as a part of Release management. Coordinate with teams like (Network, DB, Security) for the failover and patching and upgrades.
- Monitoring of applications, servers, doing capacity planning with the help of Nagios for managing logs to notify the incident management system upon reaching or exceeding the threshold limits Creating, Utilizing, Managing Policies and Glacier storage in S3 buckets on AWS.
- Coordinate release activities with Project Management, QA, Release Management and Web Development teams to ensure a smooth and trouble-free roll out of releases.
- Creating Instance in Azure, Build and Release Activities Configuration of Jenkins, Integration Activities, Deploying .net applications to Azure app service
- Deployed the applications of WAR, JAR, and EAR on multiple tomcat Server and maintained Load balancing, high availability and Failover functionality.

PSC Global solutions, Bangalore, India.
Associate Engineer.

Aug 2013-March 2015

Project:

In cite is aimed at creating web applications. Integration of fixed tasks, troubleshooting, building, releasing, configuration management deliveries etc. The SCM setup is GIT and Jenkins CI server.

Responsibilities:

- Perform health checks on servers by monitoring performance related components such as memory, CPU, network and disk activities and analyzing these reports and finding any issues that occur using Mwatch tool.
- Integration of fixed tasks, troubleshooting, building, releasing, configuration management deliveries etc.
- The SCM setup is GIT and Jenkins CI server.
- Working with developers to load software onto Linux Enterprise servers and debugging Scripts.
- Created and modified users and groups with SUDO permission.
- Working with developers to load software onto Linux Enterprise servers and debugging Scripts.
- Diagnosed and resolved problems associated with DNS, DHCP, VPN, NFS, and Apache.
- Resolved TCP/IP network access problems for the clients.
- Experience in installing and configuring SSH (Secure Shell) encryption to ensure secure access on Red Hat and Ubuntu.

EDUCATION:

- Electronics and Communication Engineering, Koneru lakshmaiah college of Engineering, bachelor of degree in 2012.